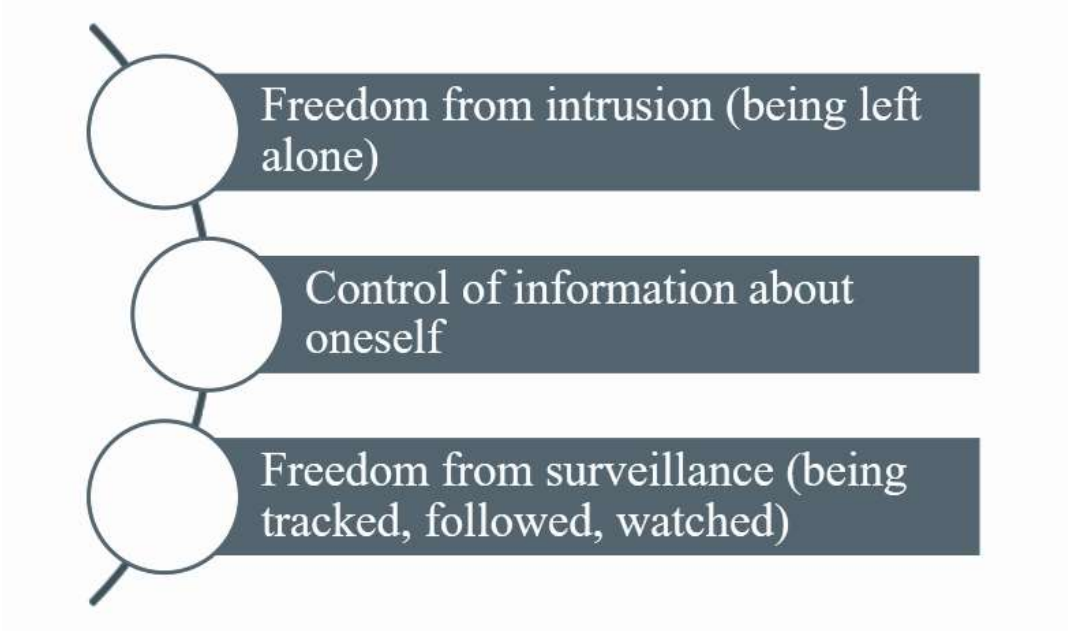
# A GIFT OF FIRE THIRD EDITION - SARA BAASE
# CHAPTER 2: PRIVACY

# Lecture 1
# Privacy and Computer Technology

Syada Tasmia Alvi
Lecturer, Dept.of CSE
Daffodil International University

# PRIVACY AND COMPUTER TECHNOLOGY

- Three Key Aspects of Privacy:



# PRIVACY RISKS AND PRINCIPLES

Privacy threats come in several categories:

- ❏ Intentional, institutional uses of personal information
- ❏ Unauthorized use or release by “insiders”
- ❏ Theft of information
- ❏ Inadvertent leakage of information
- ❏ Our own actions

# PRIVACY AND COMPUTER TECHNOLOGY

New Technology, New Risks:

- Government and private databases
  - Thousands of databases containing personal information about us.
  - Profiles of our personal information could be created easily.
- Sophisticated tools for surveillance and data analysis
  - Cameras, GPS, cell phones
- Vulnerability of data
  - Leaks of data happens, existence of data presents a risk.

# PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Examples:

- Search query data
  - Search engines collect many terabytes of data daily.
  - Data is analyzed to target advertising and develop new services.
  - Who gets to see this data? Why should we care?
- Smartphones
- Location apps
- Data sometimes stored and sent without user's knowledge

# PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- Anything we do in cyberspace is recorded.
- Huge amounts of data are stored.
- People are not aware of collection of data. Software is complex.
- Leaks happen.
- A collection of small items can provide a detailed picture.
- Re-identification has become much easier due to the quantity of information and power of data search and analysis tools.

# PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- If information is on a public Web site, it is available to everyone.
- Information on the Internet seems to last forever.
- Data collected for one purpose will find other uses.
- Government can request sensitive personal data held by businesses or organizations.
- We cannot directly protect information about ourselves. We depend upon businesses and organizations to protect it.

# PRIVACY AND COMPUTER TECHNOLOGY (CONT.)

Terminology and principles for data collection and use:

- Invisible information gathering
  - Collection of personal information about someone without the person's knowledge
  - Unauthorized software, Cookies, ISP providers •
- Secondary use
  - Use of personal information for a purpose other than the one it was provided for .
  - Data mining
    - Searching and analyzing masses of data to find patterns and develop new information or knowledge
  - Computer matching
    - Combining and comparing information from different databases (using social security number, for example, to match records)

# Computer profiling

- Using data in computer files to predict likely behaviors of people
- Analyzing data in computer files to determine characteristics of people most likely to engage in certain behavior
- Businesses find new consumers
- Government detects fraud and crime activities
- Businesses engage in profiling to determine consumer propensity toward a product or service.
- Government agencies use profiling to create descriptions of possible terrorist

# Principles for Data Collection and Use:

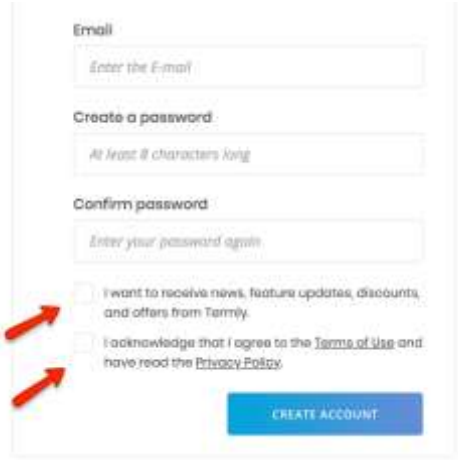
- Informed consent
- Opt-in and opt-out policies
- Fair Information Principles (or Practices)
- Data retention - Data retention is the storing of information for a specified period.

# Principles for Data Collection and Use:

- Informed consent :
  - People should be informed about the data collection and use policies of a business or organization,
  - They can then decide whether or not to interact with that business or organization.
- Opt-in and opt-out policies :
  - Opt-in :
    - Opting in means that a user will take an affirmative action to offer their consent.
    - The most common way businesses implement opt-in methods is through checkboxes. When presented with a checkbox, the user must take action to check the box, which denotes their consent.

# Principles for Data Collection and Use:

- Opt-in policies:
  - Opting in can be used in a variety of situations, including subscribing to email and newsletter mailing lists, accepting cookie use, and agreeing to legal policies.






# Principles for Data Collection and Use:

- Opt-out policies :
  - Opting out means a user takes action to withdraw their consent.
  - Not choosing to subscribe to newsletters, unticking a previously ticked checkbox, not consenting to save personal details, rejecting the use of cookies, etc. are some examples of opt-out.



# Principles for Data Collection and Use:

- Fair Information Principles (or Practices)
  - Inform people when you collect information about them, what you collect, and how you use it.
  - Collect only data needed
  - Offer a way for people to opt out from mailing lists, advertising, and other secondary uses.
  - Offer a way for people to opt out from features and services that expose personal information.
  - Keep data only as long as needed
  - Protect security of data (from theft and from accidental leaks). Provide stronger protection for sensitive data.
  - Policies for responding to law enforcement

# Thank You